Bounded Communication Fault Tolerance

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Introduction

Today:

- Bounded Communication Fault Tolerance
  1. Walktrough the math
  2. Alternative definition to Expander decomposition
  3. (d, k) - Toric
- New experimental results.
- Answering the code teleportation.

Bounded Communication Fault Tolerance

Setting

We allow a non-nosy, strong as we wish, classical computation aside. Yet we would like to limit the communication to a constant number of bits per logical qubit, in each round.

Justification

1. If we don't have a constant overhead in depth, in that model, $\Rightarrow$ we also don't have it in the standard model.

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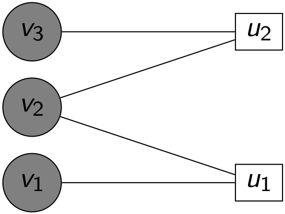
1. If we don't have a constant overhead in depth, in that model, $\Rightarrow$ we also don't have it in the standard model.
2. In that model, gate teleportation for a constant dimension code is easy. Namely, for the Topological codes the task is reduced to finding a decoding with a constant-size hint.

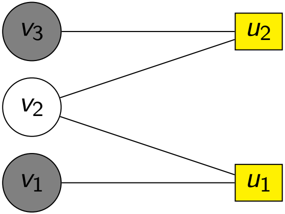
Max Color

$$\begin{tikzcd}
\text{Given - } \rho \arrow[r] \arrow[d] & \text{Decoding} \arrow[r] \arrow[d] & p\text{-depololraized} \arrow[d] \\
E_1 & E_2 & E_3
\end{tikzcd}$$

Figure: Illustration of the cycle.

v_2 is flipped. v_2 is not faulty.





If the majority of bits in the local view of v_2 are flipped, then after decoding, v_2 will be also (remain) flipped.

Strategy

Suppose we have the following Lemma:

Nice To Have Lemma.

There is a function $f : [2^n] \to \mathbb{R}$, such that for any (encoded) ρ which is subjected to q-local stochastic noise and a subset of qubits S we have:

1. With high probability $(1 - e^{-\sqrt{n}})$, after the decoding stage, the probability that the error is (partly) supported on S is bounded by $C \cdot q^{f(|S|)}$
2. $f(|S|) \geq \alpha |S|$ for some $\alpha > 0$.

Strategy

The Nice-To-Have-Lemma implies that after decoding + absorbing p-depolarized noise, we have with high probability:

$$\mathbf{Pr}\left[S\right] \leq \sum_{S' \subset S} C \cdot q^{f(S')} p^{|S/S'|} \leq C \cdot (q^{\alpha} + p)^{|S|} \leq C \cdot q^{|S|}$$

Strategy

Demonstration on the classical case.

Consider the classical expander code, defined via (γ, ε)- expander Tanner graph. Nemly, each subset on left at size at most γn sees $(1-2\varepsilon)\Delta$ uniq checks (checks that connected by single edge to the subset).

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SWIFT

What did we learn?

1. Storing time is, likily, not $\Theta(e^n)$, might be $\Theta(L^\alpha)$ for $\alpha \in (0, 1)$.
2. $\Rightarrow$ Implies that the structure of the theorem/key-lemma is different than the known refrigerators.

Future

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1. Moving forward with the experimental work?
   - More experiments, higher statistical standards?
   - Accelerate the decoder? Shifting to $c/c++$? Parallelize it with GPUs? Do we have budget?

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3. How to use the fact our code is a quotient code?